PRE-ADMISSIBILITY DESCENDS ALONG A TAUT MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement opens with the chapter's phrase 'In the situation above', which fixes the two rings and the taut map; that situation is recorded as a standard prerequisite rather than delivered as a context span.

Editorial note. The counted spans carry no navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The taut descent situation and faithful flatness of the map between the quotients. Same chapter, the situation recalled in the statement, used at proof lines 1851-1853. Allows a containment to be tested after passing to the quotients.
- Ideals of definition and weak ideals of definition of a linearly topologized ring. Same chapter, the definitions in force in the statement. Fix what has to be shown for the chosen open ideal of the source.

Lemma 0.1. *In the situation above, if B is pre-admissible, then A is pre-admissible.*

Proof. Let $J \subset B$ be a weak ideal of definition. Let $I \subset A$ be an open ideal such that $IB \subset J$. Let $I' \subset A$ be an open ideal. To show that I is an ideal of definition we have to show that $I^n \subset I'$ for $n \gg 0$. Denote $J' \subset B$ the closure of $I'B$. Then $A/I' \to B/J'$ is faithfully flat, hence injective. Thus in order to show that $I^n \subset I'$ it suffices to show that $\varphi(I)^n \subset J'$. This holds for $n \gg 0$ since $\varphi(I) \subset J$, the ideal J is an ideal of definition of B, and J' is open in B. □

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